

Session 17

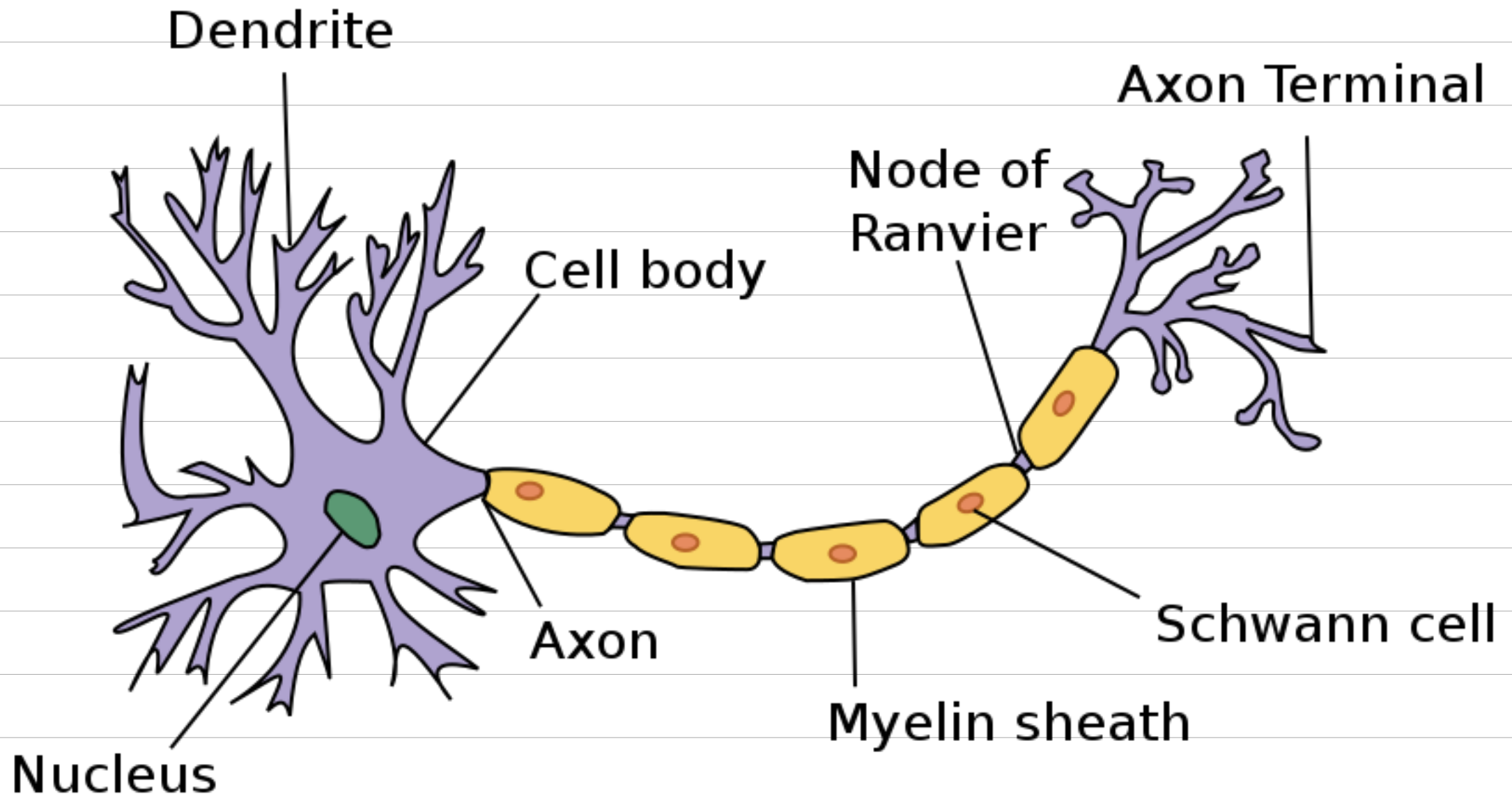
Feed Forward Neural Networks (FFNN)

Deep Learning | Zahra Amini

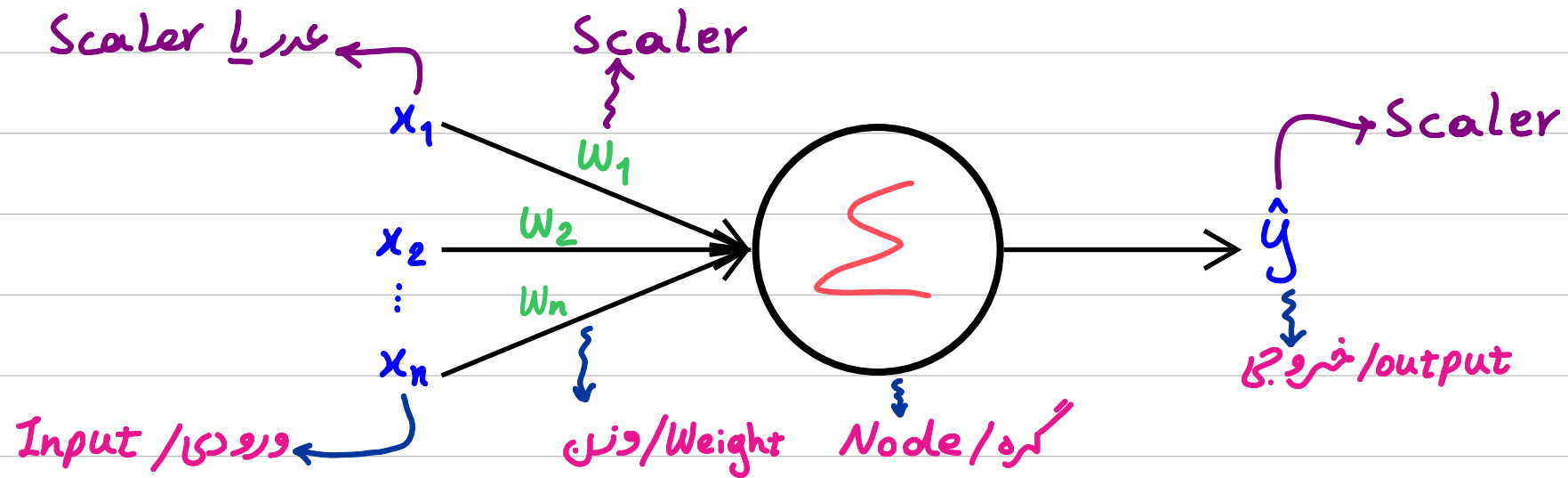
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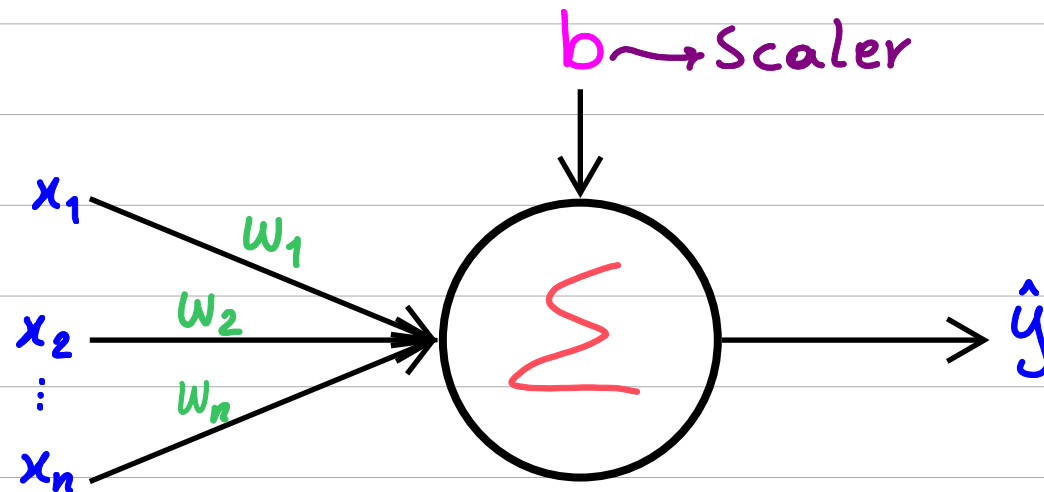
Neuron



Artificial Neuron

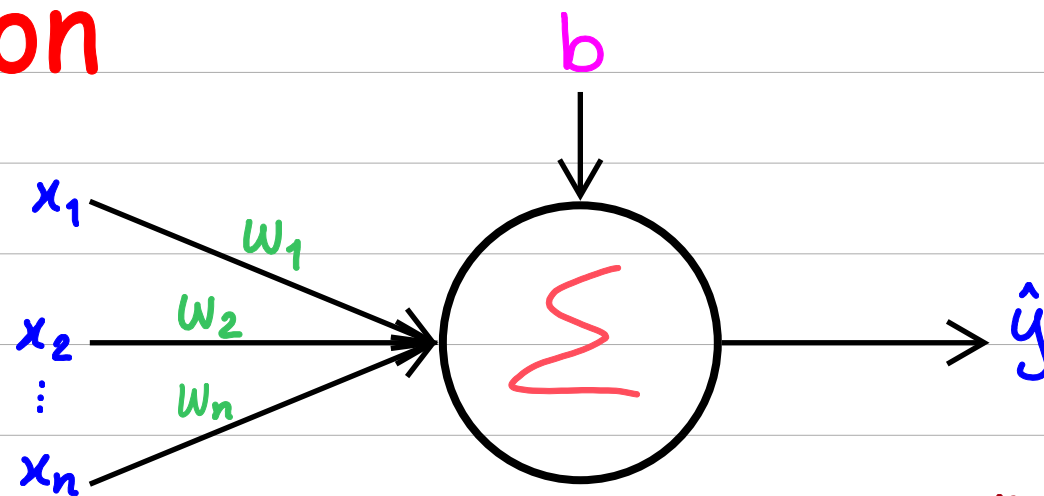


$$\hat{y} = x_1 \cdot w_1 + x_2 w_2 + \dots + x_n w_n = \sum_{i=1}^n x_i w_i$$



$$\hat{y} = x_1 \cdot w_1 + x_2 w_2 + \dots + x_n w_n + b = \sum_{i=1}^n x_i w_i + b$$

Vectorization



$$\hat{y} = x_1 \cdot w_1 + x_2 w_2 + \dots + x_n w_n + b = \sum_{i=1}^n x_i w_i + b$$

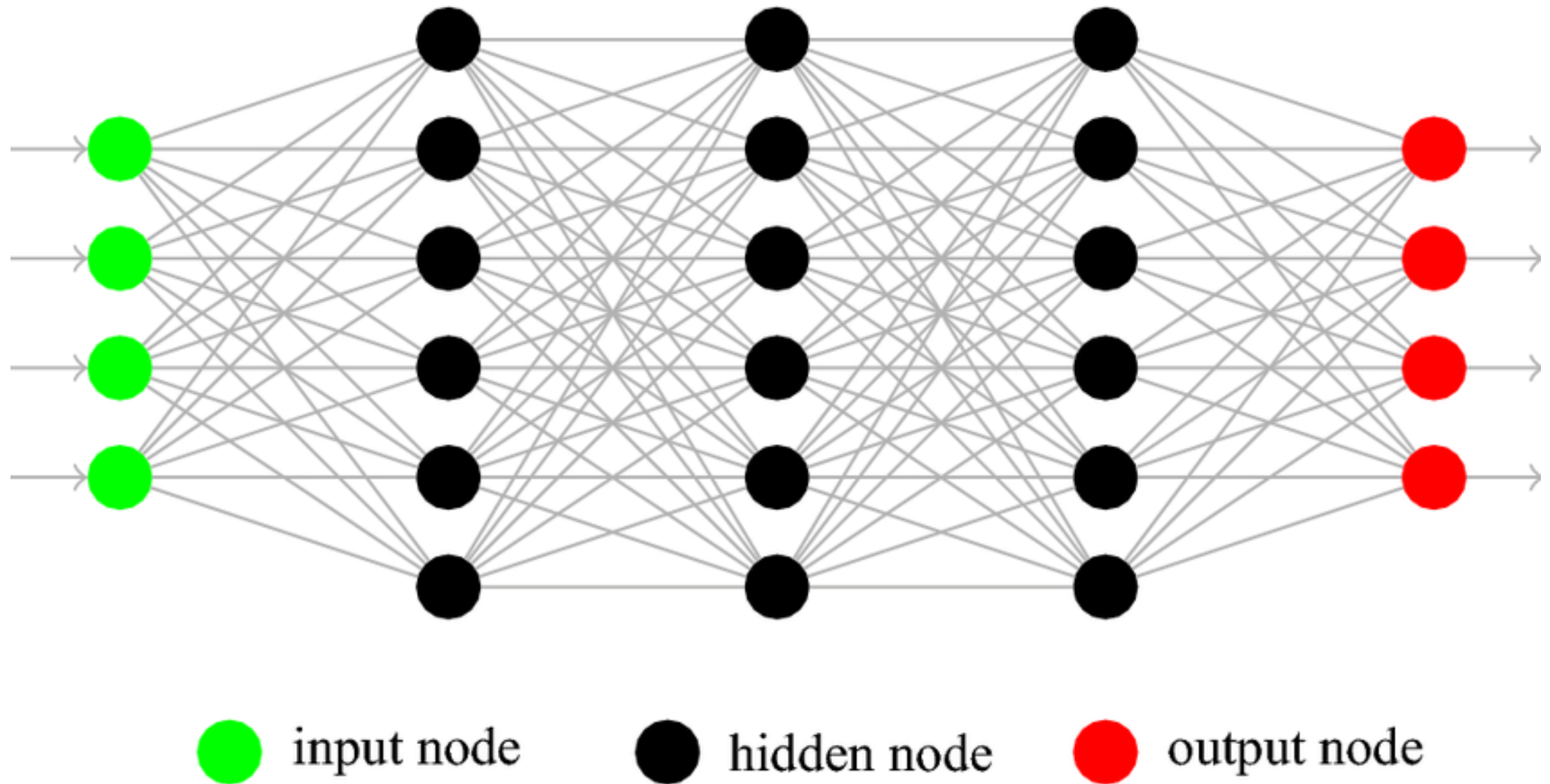
$$\begin{matrix} W \\ \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}_{n \times 1} \end{matrix} \begin{matrix} X \\ \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n \times 1} \end{matrix} + b = \hat{y} \rightsquigarrow \begin{matrix} w^T \\ [w_1 \ w_2 \ \dots \ w_n]_{1 \times n} \end{matrix} \begin{matrix} X \\ \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n \times 1} \end{matrix} + b = \hat{y}$$

✓:

$$\begin{bmatrix} a_{00} & a_{01} & \dots & a_{0n} \\ a_{10} & a_{11} & \dots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m0} & a_{m1} & \dots & a_{mn} \end{bmatrix}_{m \times n} \begin{bmatrix} a_{00} & a_{01} & \dots & a_{0p} \\ a_{10} & a_{11} & \dots & a_{1p} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n0} & a_{n1} & \dots & a_{np} \end{bmatrix}_{n \times p} = \begin{bmatrix} a_{00} & a_{01} & \dots & a_{0p} \\ a_{10} & a_{11} & \dots & a_{1p} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m0} & a_{m1} & \dots & a_{mp} \end{bmatrix}_{m \times p}$$

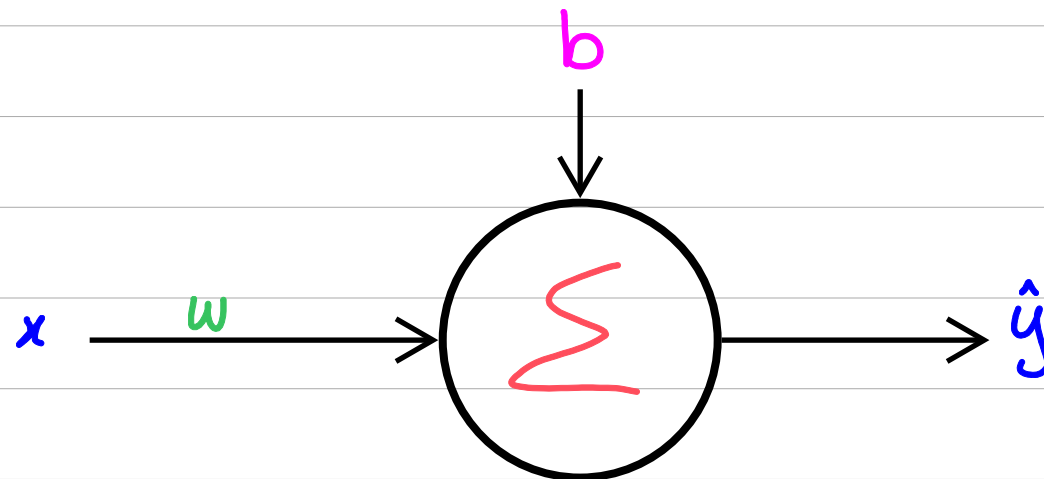
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Neural Network Layers



☑: If #Hidden layers $\geq 2 \leadsto$ Deep Neural Network

Feed-Forward Neural Network (FFNN)



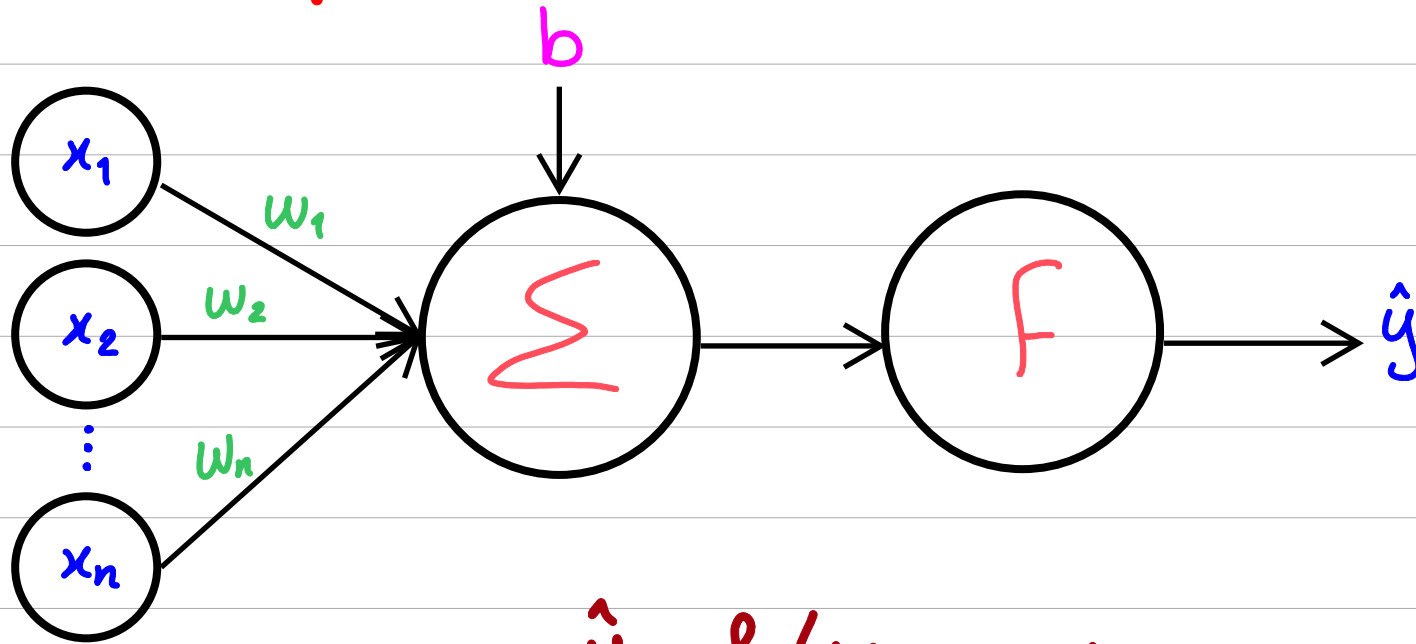
Size (m ²)	Price (k\$)	$\hat{y}$
800	200	208
1000	240	220
1200	280	232

$$w = 60 \quad b = 160000$$

$$\textcircled{1} \quad \hat{y} = wx = (60 \times 800) = 48,000 \quad 48,000 + \overbrace{160,000}^b = 208,000$$

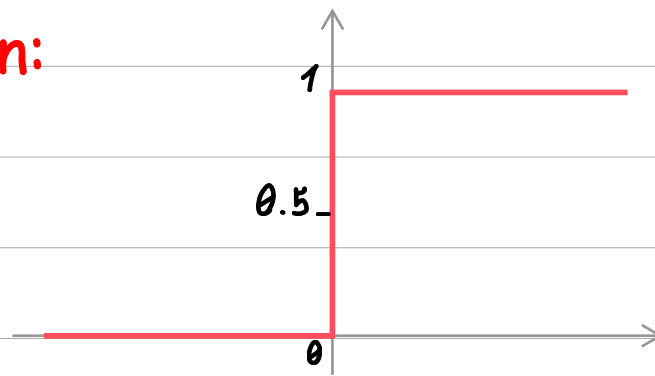
$$\textcircled{2} \quad \hat{y} = wx + b = (60 \times 1000) + 160,000 = 220,000 \quad \textcircled{3} \quad \hat{y} = wx + b = (1200 \times 60) + 160,000 = 232,000$$

Perceptron



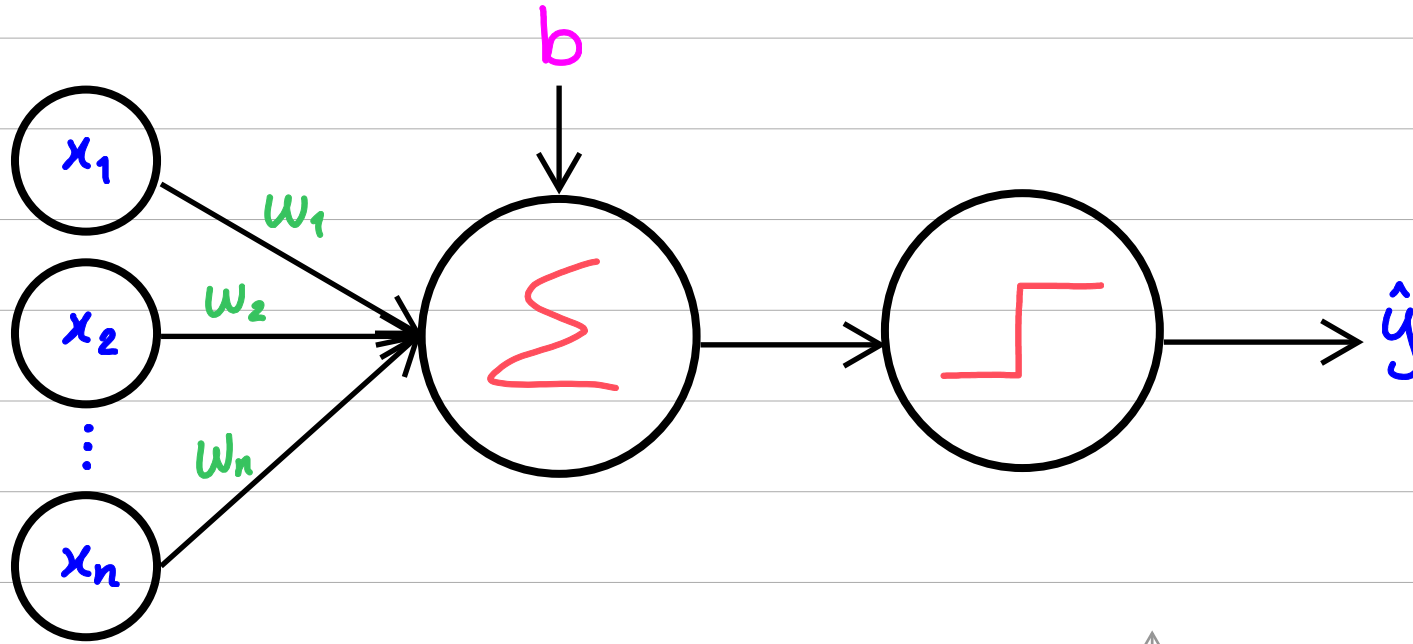
$$\hat{y} = f(w_1x_1 + w_2x_2 + \dots + w_nx_n + b) = f\left(\sum_{i=1}^n w_i x_i + b\right)$$

📌 Step Function:



$$y = \begin{cases} 1 & x > \theta \\ 0 & x < \theta \end{cases}$$

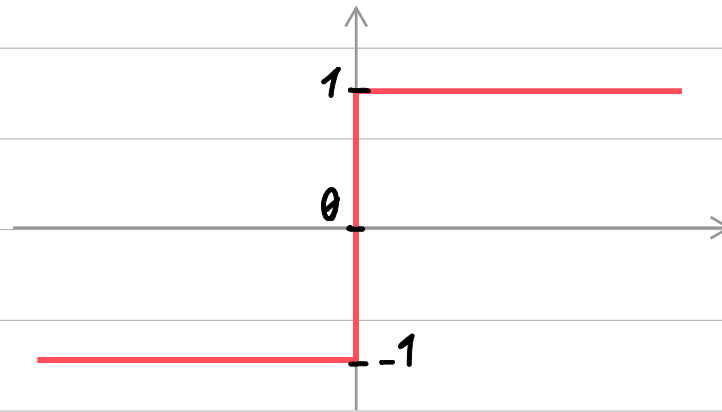
Perceptron



✂ Sign Function:



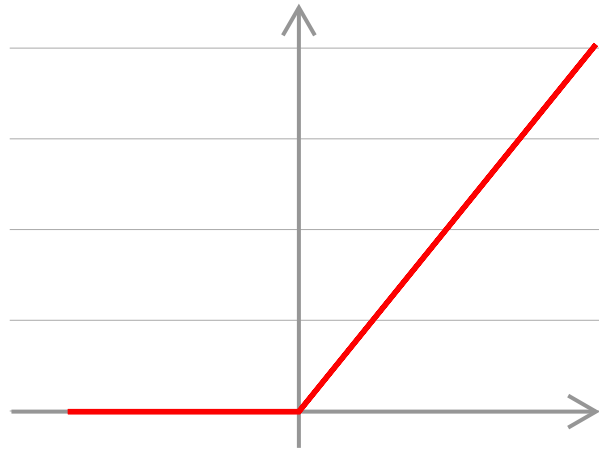
Heaviside



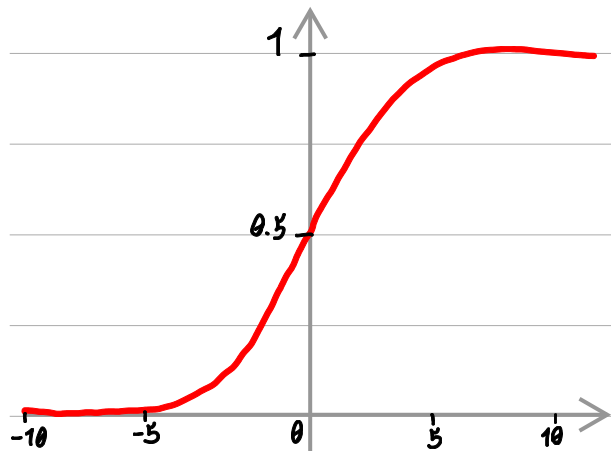
$$y = \begin{cases} 1 & x > \theta \\ -1 & \text{o.w} \end{cases}$$

$$\hat{y} = \text{Step}(w_1 x_1 + w_2 x_2 + \dots + w_n x_n + b) = \text{Step}\left(\sum_{i=1}^n w_i x_i + b\right)$$

Activation Function

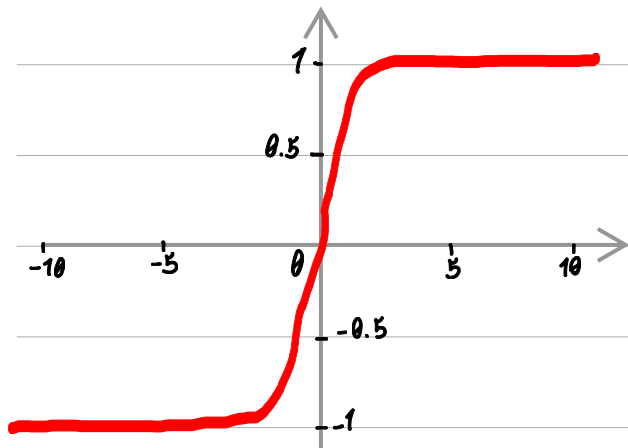


$$\text{ReLU}(z) = \max(0, z)$$



$$\text{Sigmoid} \rightsquigarrow \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\text{Softmax} \rightsquigarrow \sigma(z)_i = \frac{e^{z_i}}{\sum_{j=1}^k e^{z_j}}$$



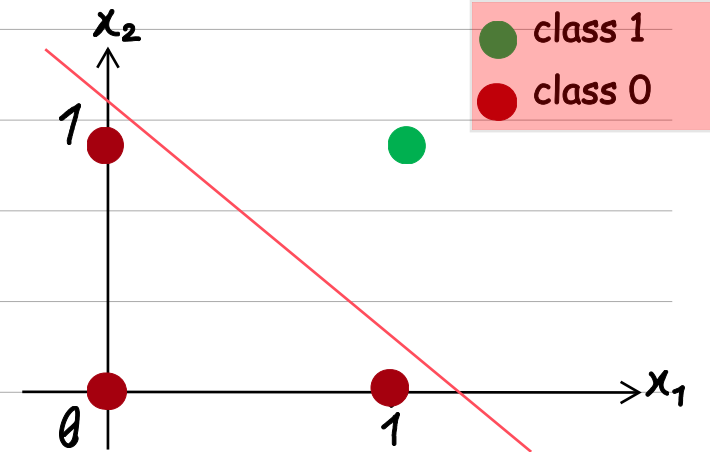
$$\text{tanh} \rightsquigarrow \sigma(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

Logic Gates

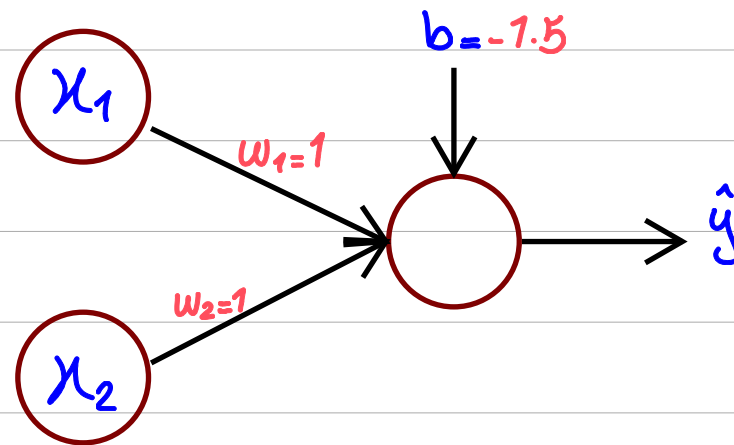
1. And &



$$\text{Step} = \begin{cases} 1 & x > 0 \\ 0 & x < 0 \end{cases}$$



x_1	x_2	$x_1 \text{ and } x_2$
T 1	T 1	T 1
F 0	T 1	F 0
T 1	F 0	F 0
F 0	F 0	F 0



$$\textcircled{1}: \hat{y} = \text{Step}(\sum w_k x_k + b) = \text{Sign}((1 \times 1) + (1 \times 1) + (-1.5)) = 1$$

$$\textcircled{2}: \hat{y} = \text{Step}((1 \times 0) + (1 \times 1) + (-1.5)) = 0$$

$$\textcircled{3}: \hat{y} = \text{Step}((1 \times 1) + (0 \times 1) + (-1.5)) = 0$$

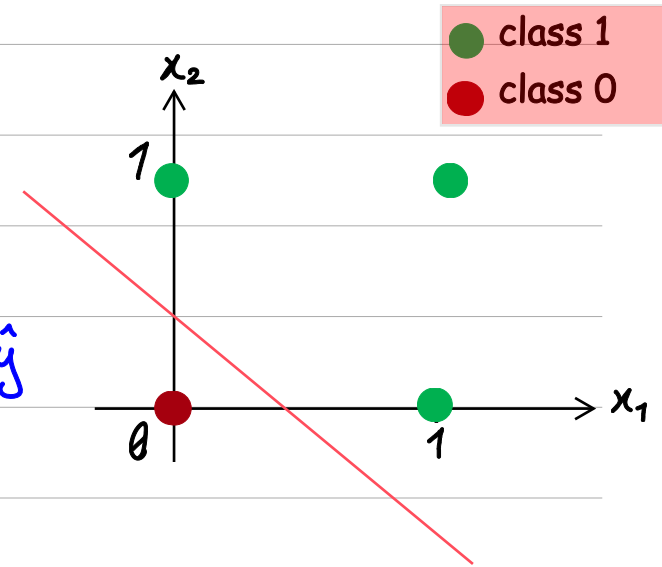
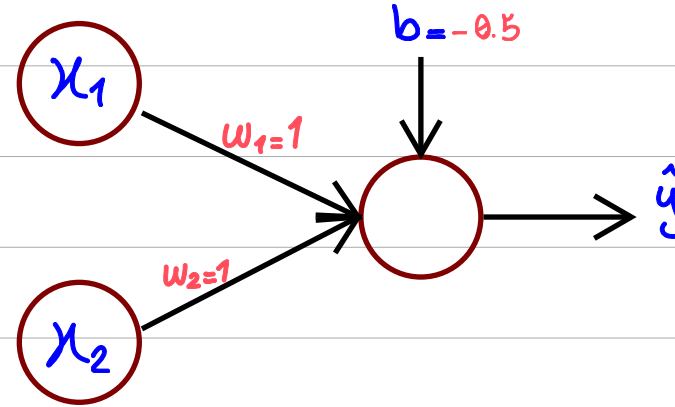
$$\textcircled{4}: \hat{y} = \text{Step}((0 \times 1) + (0 \times 1) + (-1.5)) = 0$$

Logic Gates

2. Or |



x_1	x_2	x_1 or x_2
T 1	T 1	T 1
F 0	T 1	T 1
T 1	F 0	T 1
F 0	F 0	F 0



$$\textcircled{1}: \hat{y} = \text{Step}((1 \times 1) + (1 \times 1) + (-0.5)) = 1$$

$$\textcircled{2}: \hat{y} = \text{Step}((0 \times 1) + (1 \times 1) + (-0.5)) = 1$$

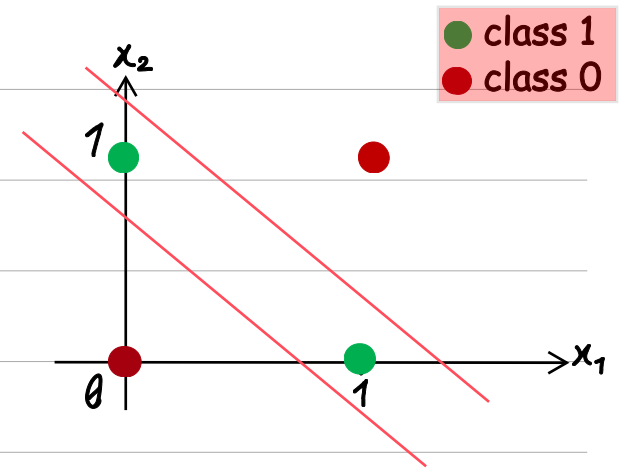
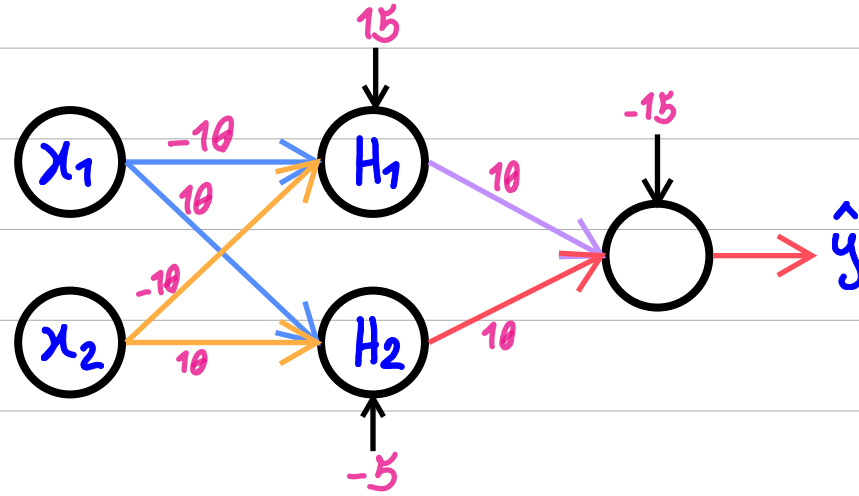
$$\textcircled{3}: \hat{y} = \text{Step}((1 \times 1) + (0 \times 1) + (-0.5)) = 1$$

$$\textcircled{4}: \hat{y} = \text{Step}((0 \times 1) + (0 \times 1) + (-0.5)) = 0$$

Multilayer Perceptron (MLP)

3. XOR

x_1	x_2	$x_1 \text{ XOR } x_2$
T 1	T 1	F 0
F 0	T 1	T 1
T 1	F 0	T 1
F 0	F 0	F 0



①. $H_1 = \text{Sigmoid}(\sum Wx + b) = \text{Sigmoid}((-10 \times 1) + (-10 \times 1) + 15) = \text{Sigmoid}(-5) = 0.006$

$H_2 = \text{Sigmoid}(\sum Wx + b) = \text{Sigmoid}((10 \times 1) + (10 \times 1) - 5) = \text{Sigmoid}(15) = 0.999$

$\hat{y} = \text{Sigmoid}((10 \times 0.006) + (10 \times 0.999) - 15) = \text{Sigmoid}(-4.88) = 0.0007 \approx 0 \rightarrow \hat{y} = 0$

②. $H_1 = \text{Sigmoid}((-10 \times 0) + (-10 \times 1) + 15) = \text{Sigmoid}(5) = 0.993$, $H_2 = \text{Sigmoid}((10 \times 0) + (10 \times 1) - 5) = \text{Sigmoid}(5) = 0.993$

$\hat{y} = \text{Sigmoid}((10 \times 0.993) + (10 \times 0.993) - 15) = \text{Sigmoid}(4.86) = 0.992 \rightarrow \hat{y} = 1$

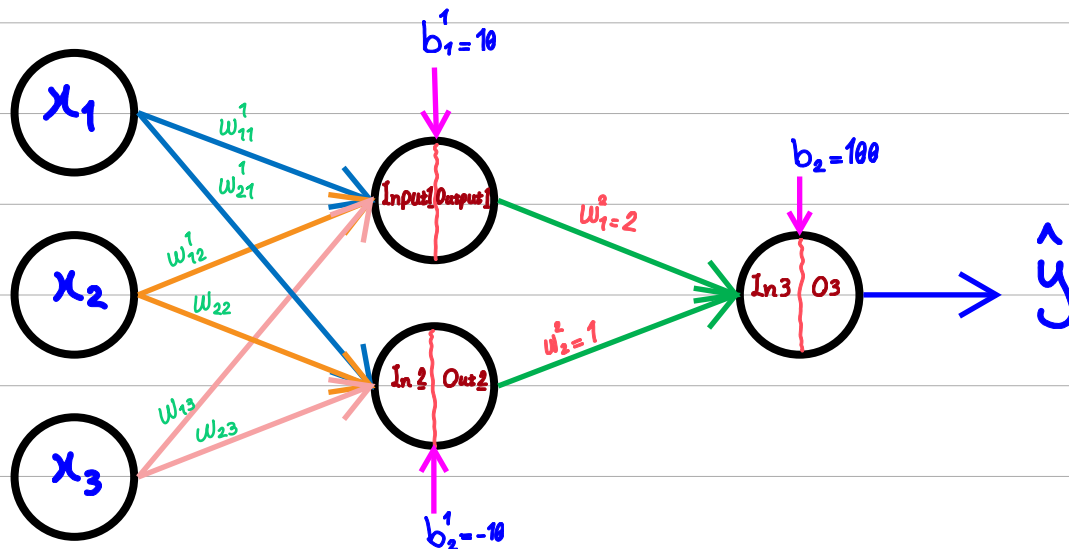
③. $\hat{y} = 1$

④. $\hat{y} = 0$

Feed-Forward Neural Network (FFNN)



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Size	#bedrooms	Age	Price
800	3	20	200

$$W_1 = \begin{bmatrix} 0.01 & 50 & 5 \\ 0.02 & 25 & -25 \end{bmatrix}, b_1 = \begin{bmatrix} 10 \\ -10 \end{bmatrix},$$

$$W_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, b_2 = 100$$

Hidden Layer

Activation Function = ReLU = $\max(0, \text{number})$

$$\text{Input 1} = \sum Wx + b = (0.01 \times 800) + (50 \times 3) + (5 \times 20) + 10 = 268 \xrightarrow{\text{AF}} \text{Output 1} = \text{ReLU}(268) = 268$$

$$\text{Input 2} = (0.02 \times 800) + (25 \times 3) + (-2.5 \times 20) + (-10) = 31 \xrightarrow{\text{AF}} \text{Output 2} = \text{ReLU}(31) = 31$$

$$\text{In 3} = (2 \times 268) + (1 \times 31) + 100 = 667 \xrightarrow{\quad} \text{Output 3} = \hat{y} = 667 \quad \text{☹}$$

Feed-Forward Neural Network (FFNN)



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☑: Normalization $\rightsquigarrow$ Value/max_value

$$\text{Size} = 800/1000 = 0.8$$

$$\text{Bedrooms} = 3/5 = 0.6$$

$$\text{Age} = 20/50 = 0.4$$

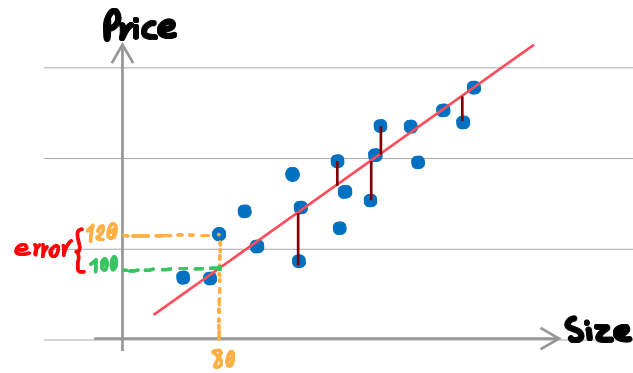
$$\text{In1} = (0.01 \times 0.8) + (50 \times 0.6) + (5 \times 0.4) + 10 = 42.008 \xrightarrow{\text{AF}} \text{Output1} = \text{ReLU}(42.008) = 42.008$$

$$\text{In2} = (0.02 \times 0.8) + (25 \times 0.6) + (-2.5 \times 0.4) + (-10) = 4.016 \xrightarrow{\text{AF}} \text{Output2} = \text{ReLU}(4.016) = 4.16$$

$$\text{In3} = (2 \times 42.008) + (1 \times 4.016) + 100 = 188.032 \rightsquigarrow \text{Output3} = \hat{y} = 188.032$$

Loss Function $\rightsquigarrow$ Error $\rightsquigarrow$

به اختلاف بین مقدار واقعی پیش بینی مدل خطای گویند.



Regression Loss

$$1. MSE = \frac{1}{2m} \sum_{i=1}^m (\hat{y}_i - y_i)^2$$

$$2. MAE = \frac{1}{m} \sum_{i=1}^m |\hat{y}_i - y_i|$$

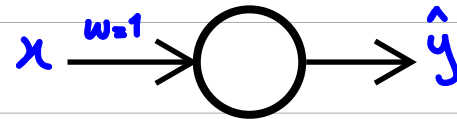
$$3. RMSE = \sqrt{MSE}$$

$$4. R^2\text{-Score} = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

SS_R (SUM of Squares For Regression)

SS_M (Sum of Squares for Model)

x	y	$\hat{y}$
1	2	1
2	4	2
3	6	3
4	8	4



$$MSE = \frac{1}{m} \sum_{i=1}^m (\hat{y}_i - y_i)^2 = \frac{1}{4} [(\cancel{1-2})^2 + (\cancel{2-4})^2 + (\cancel{3-6})^2 + (\cancel{4-8})^2] = \frac{30}{4} = 7.5$$

$$R^2 = 1 - \frac{SS_R}{SS_M} = 1 - \frac{30}{20} = -0.5 \quad ? \quad \text{منفی یا نه!}$$

$$SS_R = (2-1)^2 + (4-2)^2 + (6-3)^2 + (8-4)^2 = 30$$

$$SS_M = (2-5)^2 + (4-5)^2 + (6-5)^2 + (8-5)^2 = 20$$

$$\bar{y} = \frac{2+4+6+8}{4} = 5$$

R^2 چه عددی باشد عملکرد مدل خوبه؟

$R^2 = 1$: Perfect fit

$0 < R^2 < 1$: Good fit

$R^2 = 0$: equivalent to using the mean

$R^2 < 0$: Worse than using the mean

Thanks for
your attention